

Coded Beam Training

Dr. Vinnakota Venu Balaji

S.Harshitha Reddy - 2320040046

Gamana Chirumamilla - 2320040075

Gayathri Yerra - 2320040080

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What is Beamforming?



Definition

Beamforming is a signal processing technique that focuses the transmission or reception of signals in specific spatial directions using antenna arrays.



Purpose

Enhances signal quality, coverage, and interference suppression by aligning the phase and amplitude of antenna elements.



Intuitive Understanding

Similar to a lighthouse beam steering light, beamforming directs wireless energy toward the intended user instead of broadcasting uniformly in all directions.



Applications

Essential for 5G/6G, radar, Wi-Fi 6E, and satellite communication systems.

Types of Beams

Narrow and Wide Beams in Coded Beam Training



Narrow Beam

A highly directional beam with a small angular spread. Provides strong gain and precise alignment but requires many beams for full coverage, increasing training time.



Wide Beam

Covers a broader area with lower gain. Enables faster initial scanning across directions using fewer beams, suitable for the early stages of beam training.



Why Wide Beams Are Used in CBT

Wide beams are preferred in the initial phase because they reduce search time, offer robustness under low-SNR and mobility, and serve as coded representations of multiple narrow beams, improving efficiency and reliability.

Beam Training in XL-MIMO

- 6G systems use Extremely Large-Scale MIMO (XL-MIMO) for high spectral efficiency.
- Accurate Channel State Information (CSI) is crucial.
- Explicit CSI acquisition (channel estimation) is impractical due to:
 - ◆ Massive antennas.
 - ◆ Prohibitive pilot overhead.
 - ◆ Passive elements (e.g., XL-RIS).
- Beam training (implicit CSI acquisition) is a practical alternative:
 - ◆ Uses predefined beams (codewords).
 - ◆ Infers user direction from received power.

Key Challenge: Balancing training reliability and overhead in XL-MIMO.

Fundamental Beam Training Trade-off

- Beam training faces a core trade-off: Reliability (works for low-SNR, cell-edge users) vs. Efficiency (low overhead).
- Existing methods typically optimize only one:
 - ◆ High reliability -> high overhead
 - ◆ Low overhead -> fragile to noise
- This trade-off worsens in XL-MIMO due to:
 - ◆ Wider coverage
 - ◆ More remote users
 - ◆ Strong SNR imbalance

What, Why, and How of Codebook Beamforming



What is Codebook Beamforming?

A pre-defined set of beam patterns (codebook) is used for beam selection, reducing search complexity and training overhead.



Why Use a Codebook?

It enables efficient beam alignment with minimal feedback, standardized across devices and base stations for interoperability.



How It Works

The transmitter and receiver share a known codebook; during training, the best beam index is selected based on signal strength feedback.



Advantages

Fast, scalable, and suitable for multi-user MIMO systems; allows adaptive beam switching with low latency.

Our Explanation of Codebook Beamforming



Coded Logic

CBT uses error-correcting codes (e.g., Hamming codes) to represent beam directions. Each beam index is mapped to a binary codeword for robust identification.



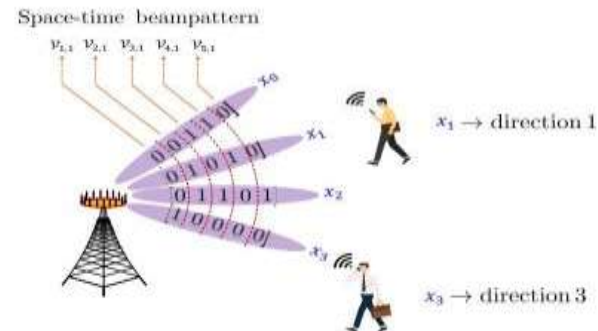
Decoding Process

An error-correcting decoder identifies the true beam index, compensating for noisy or missed measurements.



Measurement and Feedback

The user measures signal strengths for a coded sequence of beams and reports binary feedback patterns to the base station.



Why We Use Wide Beams

Efficiency and Robustness in Beam Training

- **Fast Initial Search:** Wide beams cover large angular regions, enabling quick user direction discovery without scanning each narrow beam.
- **Reduced Training Overhead:** Minimizes the number of training slots needed, conserving pilot power and time resources.
- **Robustness Under Low SNR:** Wide beams maintain detection accuracy even in noisy or interference-heavy environments.
- **Mobility and Adaptation:** Ideal for fast-moving users, as wide beams allow rapid beam realignment in dynamic conditions.
- **Role in Coded Beam Training:** In CBT, wide beams encode multiple narrow beams, combining speed, reliability, and efficient decoding.

Aspect	Hamming (Block)	Convolutional
Use	Perfect (10 fixed Tx → 64 identify)	Streaming (not fixed beams)
SNR	$t=1$ correction @ -15dB	Viterbi complex for low SNR
Overhead	10 beams = optimal	Rate 1/2 = 12 beams needed
Decoding	Simple syndrome ($O(1)$)	Viterbi trellis ($O(2^K)$)
XL-MIMO	Fixed beam search	Not standard

PLOTS

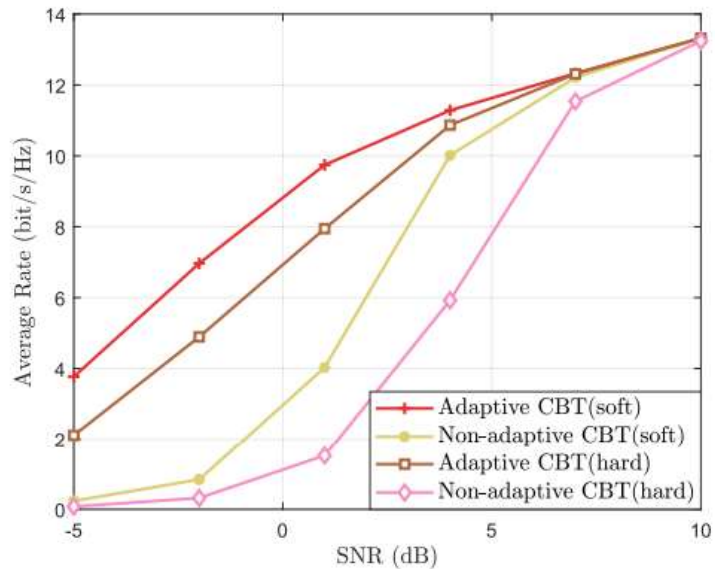


Fig. 1. Comparisons of the average rate for adaptive/non-adaptive CBT with soft/hard decoders.

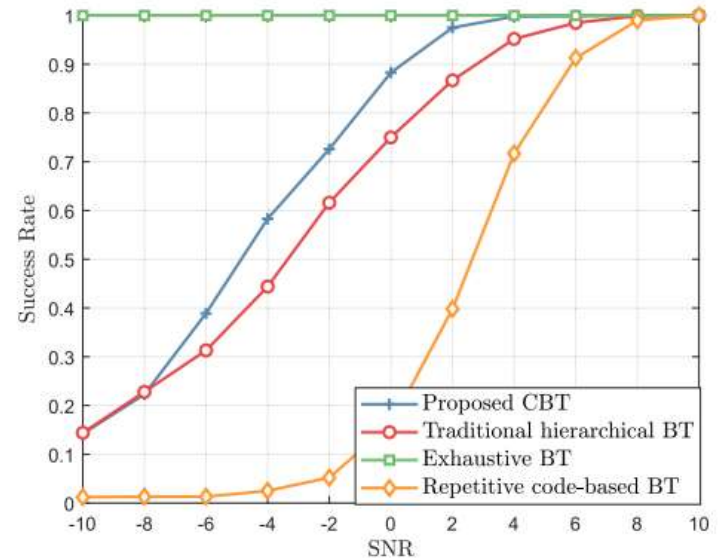


Fig. 2. Comparisons of the success rate for different beam training methods.

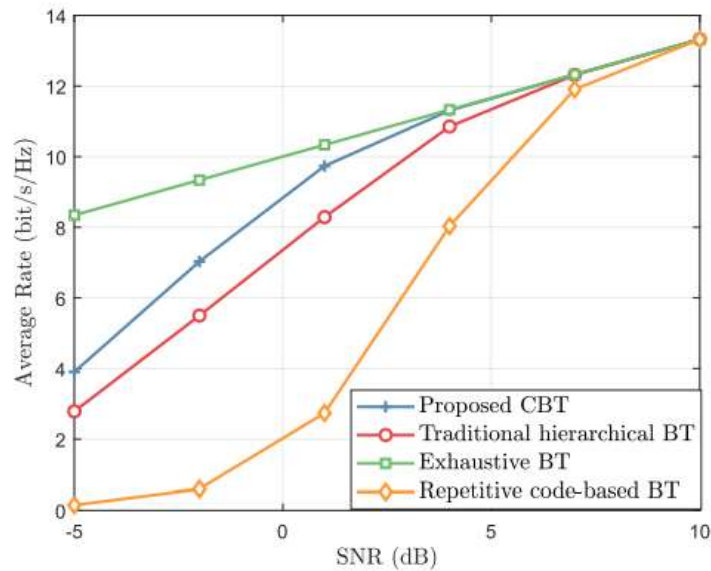


Fig. 3. Comparisons of the average rate for different beam training methods.

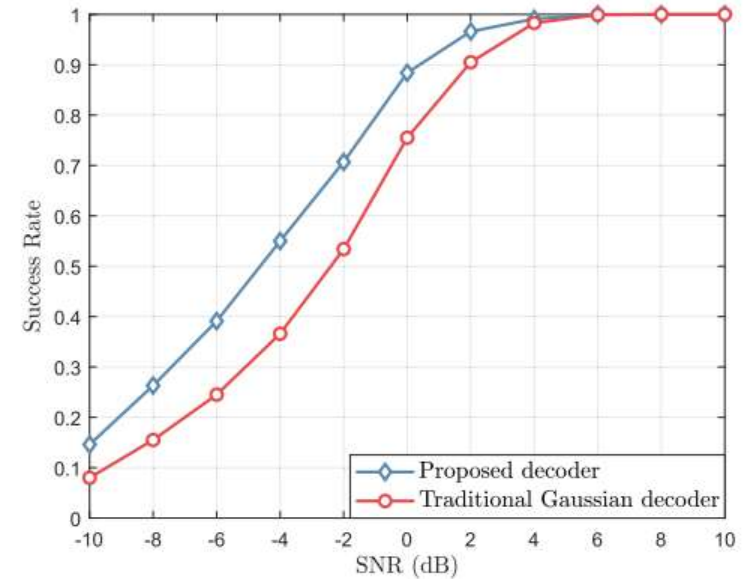


Fig. 4. Comparisons of the performance for convolutional decoders with different LLR calculators.

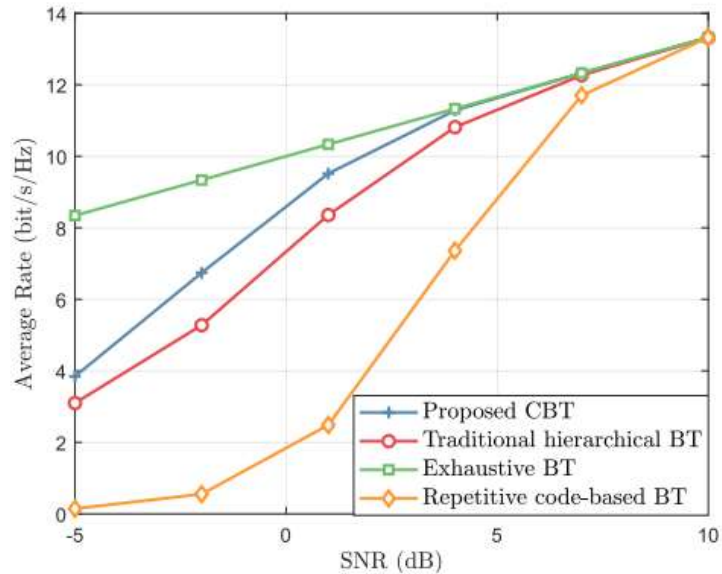


Fig. 5. Comparisons of the average rate for different beam training methods in mixed LoS and NLoS channel.

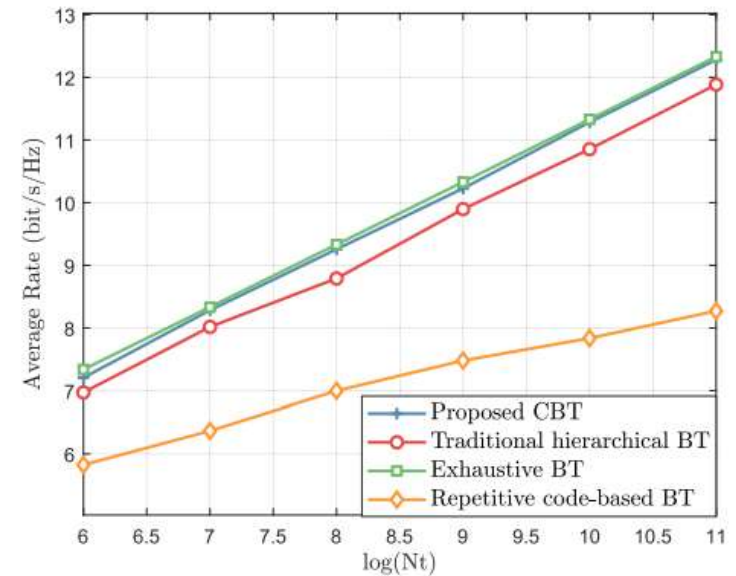


Fig. 6. Average rate performance under different BS antenna configurations

THANK YOU